

Daniel L. Perret, PhD

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SUMMARY

- **Forest ecologist and conservation biogeographer** with expertise in forest ecosystem responses to global change, ecological theory and concepts, and climate and disturbance ecology
- Demonstrated success in project management, science communication, and timely production of high-impact deliverables
- Motivated to pair creative science with meaningful collaborations to build the knowledge and partnerships we need to confront urgent problems in forest management and conservation

EDUCATION

- PhD, Brown University** 2022
Niche-based perspectives on species' responses to climate change: pines as a multi-scale case study
Department of Ecology, Evolutionary & Organismal Biology
Advisor: Prof. Dov F. Sax
- BS, Stanford University** 2013
Biology (Ecology and Evolutionary Biology, *with Honors*)
Advisor: Prof. Elizabeth A. Hadly

CORE COMPETENCIES

Forest ecology & Biogeography

Ecosystem reorganization dynamics · Demography and population biology · Landscape and disturbance ecology · Global change biology · Forest Inventory & Analysis (FIA)

Data Analysis & Statistics

Bayesian hierarchical methods · Geospatial analyses (GEE, ArcGIS, raster and vector data) · Ecological modelling and forecasting · Data management and analysis in R · Version control (Git/GitHub)

Science Communication

Technical and general-audience writing · Creative data visualization · Workshops & symposiums · Leading peer-reviewed publications

Project Management

Interdisciplinary collaboration · Grant proposal development · Timelines and deliverable production · Stakeholder engagement

PROFESSIONAL EXPERIENCE

- Postdoctoral Research Fellow** 2022 - present
United States Forest Service Pacific Northwest Research Station (ORISE)
- Lead researcher on projects quantifying forest resilience to climate and disturbance using forest inventories, tree-ring data, field surveys, remote sensing data, and novel Bayesian hierarchical models
 - Work garnered invited commentaries by international scientists in *PNAS* and *Global Change Biology*
 - Co-led development of >\$1M in competitive proposals for new forest resilience research
- Graduate Research Assistant** 2016 - 2022
Department of Ecology, Evolution, & Organismal Biology, Brown University

- Developed multi-scale assessments of pine species' climate vulnerability using ecological niche models, tree-ring data, and field surveys
- Published findings on ecological forecasting of climate vulnerability in *PNAS*, *Ecography*, *Nature Climate Change*, and *Global Ecology & Biogeography*
- Led extensive field campaigns internationally and across the western United States

Spatial Analyst, University of Montana

2014 – 2016

- Developed and deployed spatiotemporal wetland distribution models at landscape scales for Greater Sage Grouse conservation across the western US
- Collaborative interagency project resulted in interactive web platform for management planning and decision-making

Field Research Technician, Yellowstone Wolf Project

2013 – 2014

- Extensive backcountry fieldwork in support of high-profile wolf and mountain lion research

PEER-REVIEWED PUBLICATIONS

** published with invited external commentary*

Evans, M. E. K., P. B. Adler, A. L. Angert, S. M. N. Dey, M. P. Girardin, K. A. Heilman, S. Klesse, **D. L. Perret**, D. F. Sax, S. N. Sheth, M. Stemkovski, J. L. Williams. 2025. Reconsidering space-for-time substitution in climate change ecology. *Nature Climate Change*. <https://doi.org/10.1038/s41558-025-02392-0>

***Perret, D. L.**, D. M. Bell, H. S. J. Zald. 2025. Reducing fire severity and extent bolsters subalpine forest resilience to global change through key demographic pathways. *Global Change Biology*. <https://doi.org/10.1111/gcb.70052>

Kissling, M. L., M. C. McDevitt, J. Felis, **D. L. Perret**, P. M. Lukacs, J. Johnson, M. Mazonna. 2025. Nesting ecology and movements of an ice-associated seabird, the Kittlitz's murrelet (*Brachyramphus brevirostris*) at the northern edge of its range. *Wildlife Society Bulletin*. <https://doi.org/10.1002/wsb.1575>

***Perret, D. L.**, M. E. K. Evans, D. F. Sax. 2024. A species' response to spatial climatic variation does not predict its response to climate change. *PNAS*. <https://doi.org/10.1073/pnas.2304404120>

Perret, D. L., D. M. Bell, A. N. Gray, J. D. Shaw, H. S. J. Zald. 2023. Range-wide population assessments for subalpine fir indicate widespread disturbance-driven decline. *Forest Ecology & Management*. <https://doi.org/10.1016/j.foreco.2023.121128>

Perret, D. L., & D. F. Sax. 2021. Evaluating alternative study designs for optimal sampling of species' climatic niches. *Ecography*. <https://doi.org/10.1111/ecog.06014>

Rosenblad, K., **D. L. Perret**, D. F. Sax. 2019. Niche syndromes reveal climate-driven extinction threat to island endemic conifers. *Nature Climate Change*. <https://doi.org/10.1038/s41558-019-0530-9>

Perret, D. L., A. B. Leslie, D. F. Sax. 2019. Naturalized distributions show that climatic disequilibrium is structured by niche size in pines (*Pinus* L.). *Global Ecology & Biogeography*. <https://doi.org/10.1111/geb.12862>

Donnelly, P. J., B. W. Allred, **D. L. Perret**, N. L. Silverman, J. D. Tack, V. J. Dreitz, & D. E. Naugle. 2019. Seasonal drought in North America's sagebrush biome structures dynamic ecological minimums for sage-grouse. *Ecology & Evolution*. <https://doi.org/10.1002/ece3.4614>

MANUSCRIPTS IN PROGRESS

McLaughlin, B. P., M. Kling, S. N. Miller, **D. L. Perret**, B. J. McGill. An ecology-focused, new-user introduction to the Forest Inventory and Analysis (FIA) Database.

Perret, D. L., D. M. Bell, H. S. J. Zald. Plots, planes, pixels: multi-scale data integration for dynamic monitoring of forest mortality and risk from insects, diseases, and drought.

Perret, D. L., D. M. Bell, H. S. H. Zald. Dendroecological indicators of sugar pine (*Pinus lambertiana*) vulnerability and decline across its geographic and climatic distribution.

GRANTS & AWARDS

- Plots, planes, & pixels: multiscale data integration for dynamic monitoring of forest mortality and risk from insects, diseases, and drought. (submitted 2025)* (\$824,622)
Science PI – **Daniel Perret**, PI – Harold Zald, Co-I – David Bell
NASA ROSES-24 A.60 Ecological Conservation grant proposal supporting data integration and fusion to support mapping and estimation of forest mortality and future vulnerability to biological disturbances.
- Plots, planes, & pixels: biological disturbance agent estimation. (2023)* \$114,000
PI – Harold Zald, Co-PIs – David Bell, **Daniel Perret**
ORISE Fellowship supported by interagency agreement between the US Forest Service Pacific Northwest Research Station and the US Department of Energy.
- Vulnerability assessment of sugar pine, an iconic species in decline. (2022)* \$100,000
PI – Harold Zald, Co-PIs – David Bell, Andrew Gray, **Daniel Perret**
ORISE Fellowship supported by interagency agreement between the US Forest Service Pacific Northwest Research Station and the US Department of Energy.
- Does ecological niche theory predict tree performance in exotic communities and climates? (2020)* \$9,935
Doctoral Dissertation Enhancement Grant, The Bushnell Graduate Research & Education Fund.
- Reading between the rings: the climate future of North American pines. (2019)* \$4,596
Graduate Research, Travel & Training Grant, Institute at Brown for Environment and Society.
- Assessing climate-driven extinction risk by measuring performance of North American pines. (2018)* \$4,000
Lewis and Clark Fund for Exploration and Field Research, The American Philosophical Society.
- Assessing climate-driven extinction risk by measuring the population performance of pines. (2018)* \$4,977
Graduate Research, Travel & Training Grant, Institute at Brown for Environment and Society.
- Spatial distribution and morphological variation in the mountain beaver (*Aplodontia rufa*). (2011)* \$2,600
Stanford University Undergraduate Advising and Research Small Grant.

ORGANIZED SYMPOSIA

- Demographic perspectives on forest responses to global change.* Lead organizer for symposium at the US Forest Service Forest Inventory and Analysis Science Symposium. November 2024.
Speakers: Ayanna St. Rose (OSU), Yannek Käber (), Kyle Rosenblad (UC Berkeley), Adam Noel (USGS), Margaret Evans (UArizona), **Daniel Perret** (USFS)
- Forest Inventory Data for Global Change Ecology: Opportunities, Challenges, and Innovations.* Lead organizer for symposium at Ecological Society of America. Portland, OR. August 2023.
Speakers: Sara Goeking (USFS), Bailey McLaughlin (UMaine), Erin Berryman (USFS), Margaret Evans (UArizona), Robert Shriver (UNevada, Reno), Sarah Jovan (USFS)
- An Introduction to Estimating Ecological Change with the US Forest Service Forest Inventory and Analysis (FIA) Database.* Lead organizer for workshop at Ecological Society of America. Portland, OR. August 2023.
Instructors: **Daniel Perret** (USFS), Harold Zald (USFS), Jane Terzibashian (USFS), Bailey McLaughlin (UMaine)

INVITED TALKS

- Spatial and temporal dimensions of forest resilience to global change: evidence from tree rings and forest inventories.* Invited seminar, University of Arizona Laboratory of Tree Ring Research. September 2025 (planned).
- Plots, planes, and pixels: comparing and integrating datasets to improve forest mortality detection.* US Forest Service FIA Science Symposium. November 2024.

Changing world, changing forests: global, continental, and regional perspectives. Invited seminar, University of Montana. Missoula, MT. April 2024.
Interactions between climate and disturbance are driving widespread subalpine fir decline. FIA Stakeholder Science Meeting. November 2022.

CONTRIBUTED TALKS & POSTERS

Reducing fire severity and extent bolsters subalpine forest resilience to global change through key demographic pathways. US Forest Service FIA Science Symposium. November 2024.
Range-wide population assessments for subalpine fir: widespread disturbance-driven decline. USFS Pacific Northwest FIA Operational Meeting. January 2024.
Interactions between climate and disturbance are driving widespread subalpine fir decline. MtnClimate. Crested Butte, CO. September 2022.
Population-specific responses determine species' vulnerability to climate change: a spatiotemporal case study of ponderosa pine. Ecological Society of America. Montreal, Canada. August 2022.
Tree ring records reveal that climate envelope approaches underestimate population vulnerability to climate change. International Biogeography Society. Vancouver, Canada. January 2022.
Niches, tree-rings, and invasive forests. Brown University EEOB Departmental Seminar. Dec 2020.
Tree rings and niche models as tools for predicting forest responses to climate change. Earth, Itself. Brown University. March 2019. (poster)
Is all niche space created equal? A framework for exploring the structure of the niche. International Biogeography Society. Malaga, Spain. Jan 2019. (poster)
Climatic disequilibrium is structured by niche size. International Biogeography Society, Climate Change Biogeography. Evora, Portugal. March 2018.
Niche size structures climatic disequilibrium in pines (Pinus L.). New York Species Distribution Modeling Symposium. Yale University. December 2017.

MEDIA & OUTREACH

Forest health, disturbance, and resilience in the Northern Rockies and the Blackfoot watershed. Aspen Ecology Workshop, Greenough MT. July 2025.
Mountain face off: How trees choose sides. "The Big Why", Montana Public Radio. July 2025.

TEACHING

Conservation Biology (Biol 1470) , teaching assistant <i>Brown University</i>	2020
Guest lectures: <i>Wildlife Management; Wildlife, Climate, and Conservation</i>	
Conservation Biology (Biol 1470) , teaching assistant <i>Brown University</i>	2017

PROFESSIONAL WORKSHOPS

Dendrochronology Intensive Summer Course Laboratory of Tree-Ring Research, University of Arizona	2018
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PROFESSIONAL SERVICE

Reviewer for *PNAS, Science Advances, New Phytologist, Ecography, Global Ecology & Biogeography, Ecosphere, Ecology & Evolution, Forest Ecology & Management*

REFERENCES

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Pacific Northwest Research Station

USDA Forest Service

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